Date: 2080/12/21

Class: BSc 2nd Year

Roll No: 68

Shift: Day

Physics Practical Sheet

Prithvi Narayan Campus

Experiment No: 6 (A)

Group: P_1

Sub: Physics

Set: 1

Object of the Experiment (Block Letters): TO STUDY NAND GATE AS AN UNIVERSAL GATE BY USING IC 7400.

APPARATUS REQUIRED:

(i) Diode (ii) Wires (iii) Transistor
(iv) Multimeter (v) Bread board (vi) IC (7400)

THEORY:

The NAND gate can be used to perform all the three logic functions of OR, NOT and AND gates. This is why, it is known as universal gate.

(I) OR gate:

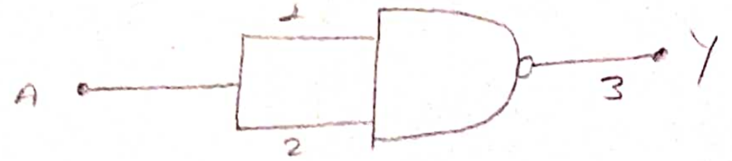
As shown in the figure, an OR gate can be made out of three NAND gates. The inputs A and B are inverted by two NAND gates, The before they are applied to the last NAND gate. The output is; $Y = \overline{\bar{A}\bar{B}}$; which is equal to $Y = A+B$ according to De-Morgan's theorem.

(II) AND gate:

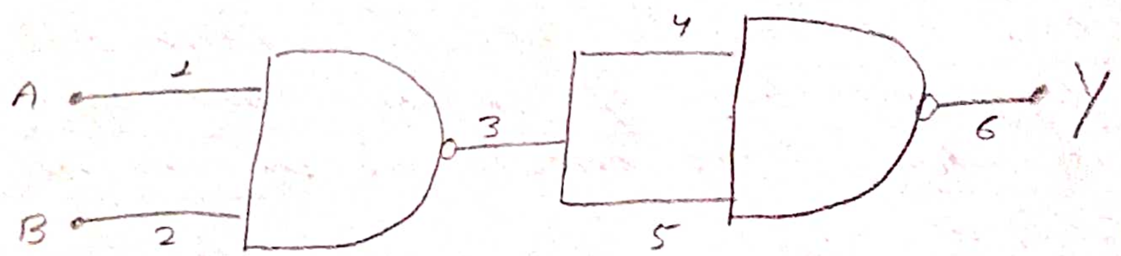
As shown in figure, ~~an~~ AND gate can be made out of two NAND gates. The output is $Y = \overline{\overline{A \cdot B}}$ which is equal to $A \cdot B$. i.e, $Y = A \cdot B$.

(III) NOT gate:

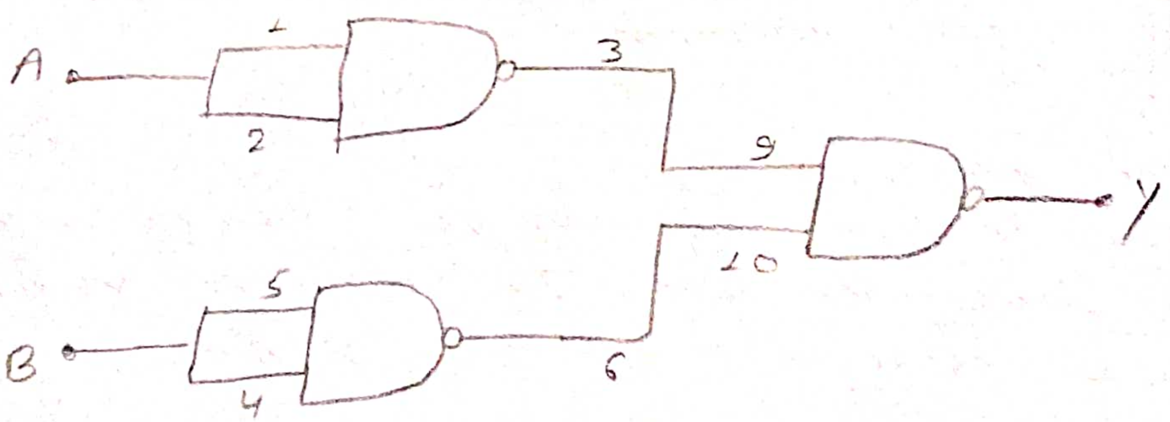
As shown in the figure, a NOT gate can be made out of NAND gate by connecting its two inputs together. When a NAND gate is used as a NOT gate, the other logic symbol is employed. The output is $Y = \overline{AA}$ i.e. $Y = \bar{A}$.



(I) NAND as NOT gate



(II) NAND as AND gate



(III) NAND as OR gate

Fig: NAND Gate as Universal gate

OBSERVATION:

(I) OR gate

S.N	Inputs		Output
	A	B	Y = A+B
1.	0	0	0.03
2.	0	4.49	4.56
3.	4.49	0	4.47
4.	4.49	4.49	4.60

(II) AND gate

S.N	Inputs		Output
	A	B	Y = A.B
1.	0	0	0
2.	0	4.41	0
3.	4.41	0	0.01
4.	4.41	4.41	4.40

(III) NOT gate

S.N	Input	Output
	A	Y
1.	0	4.37
2.	4.31	0.07

RESULT:

Hence, the NAND gate is verified as universal gate by using IC 7400.

PRECAUTION:

(i) The current through the IC should be suitable for the I.C. so, that it could not fuse.

(ii) The reading and connection should be done carefuly.